

Statistics Practice Set — S18

S18 will push you one step beyond S17: **business interpretation + choosing the right statistic + comparing distributions.**

No basic Population/Sample question and no formula-recall question.

Q1 — Business Decision: Mean vs Median

A company records employee annual bonuses:

₹18,000, ₹20,000, ₹22,000, ₹24,000, ₹25,000, ₹27,000, ₹1,50,000

Answer:

1. Calculate the mean.
 2. Calculate the median.
 3. Which measure should HR use to represent the **typical employee bonus**?
 4. Explain why.
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Q2 — Standard Deviation vs CV

Two stores have the following sales data:

Store	Mean Sales	SD
A	₹30,000	₹3,000
B	₹90,000	₹6,000

Answer:

1. Calculate CV for both.
2. Which store has greater **absolute variability**?
3. Which store has greater **relative variability**?
4. Which store is more consistent relative to its sales level?
5. Why would looking only at SD be misleading?

Q3 — IQR and Outlier Resistance

Dataset A:

10, 12, 14, 15, 16, 18, 20

Dataset B:

10, 12, 14, 15, 16, 18, 100

Answer:

1. Which dataset is more affected by the extreme value?
2. Which measure would be better for comparing the spread of the typical observations: **Range or IQR?**
3. Explain why.
4. What happens to the range when 20 is replaced by 100?
5. Would the IQR be affected as strongly? Why?

Q4 — Distribution Shape

A dataset has:

Statistic	Value
Minimum	8
Q1	15
Median	20
Q3	32
Maximum	70

Answer:

1. Calculate the lower whisker length.
 2. Calculate the upper whisker length.
 3. Which side has the longer tail?
 4. Is the distribution likely left-skewed, right-skewed, or approximately symmetric?
 5. Explain your conclusion using **two pieces of evidence**.
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Q5 — Box Plot Comparison

Statistic	Product A	Product B
Minimum	20	10
Q1	30	25
Median	40	35
Q3	50	60
Maximum	65	100

Answer:

1. Which product has the higher median?
 2. Calculate the IQR of both.
 3. Calculate the range of both.
 4. Which product has greater variability?
 5. Which product appears more consistent?
 6. Which product has the longer upper tail?
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Q6 — CV Ranking

Three investment options have:

Investment	Mean Return	SD
A	8%	1.6%
B	15%	3%
C	25%	4%

Calculate the CV for each.

Then:

1. Rank them from **most consistent** → **least consistent**.
 2. Which has the lowest relative variability?
 3. Does the investment with the highest return also have the lowest relative variability?
 4. Explain.
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Q7 — Outlier Detection

Dataset:

12, 14, 15, 17, 18, 20, 22, 24, 25, 55

Using the IQR method:

1. Find Q1.
2. Find Q3.
3. Find IQR.
4. Find the lower fence.
5. Find the upper fence.
6. Identify the outlier, if any.
7. Explain why it is an outlier.

Q8 — Choose the Correct Measure

Choose the **best option** for each situation.

Options:

- Mean
- Median
- SD
- IQR
- CV
- Box plot

A.

A CEO wants to know the typical salary of employees when a few executives earn extremely high salaries.

B.

An analyst wants to compare relative variability between two products with very different average prices.

C.

An analyst wants to measure the spread of the middle 50% of observations.

D.

An analyst wants to compare five departments' distributions and identify possible outliers visually.

E.

A company wants to know the absolute amount of variation in delivery times when both teams have the same average delivery time.

F.

An analyst wants to calculate the overall arithmetic average of a dataset without major extreme values.

Q9 — Find the Wrong Interpretation

A junior analyst says:

“Company A has a standard deviation of ₹5,000 and Company B has a standard deviation of ₹8,000. Therefore, Company A is definitely more consistent.”

Is this conclusion always valid?

Answer:

1. Yes or No?
2. Explain.
3. Give an example where the company with the **higher SD** actually has the **lower CV**.

Q10 — Full Analyst Case

Two delivery companies report:

Company	Mean	Median	SD
A	35 min	30 min	7 min
B	60 min	59 min	9 min

Answer:

1. Calculate CV for both companies.
2. Which company has lower absolute variability?
3. Which company has lower relative variability?
4. Which company appears more right-skewed?
5. Which company would you choose if the goal is **relative consistency**?
6. Which company would you choose if the goal is **lower absolute variation**?
7. Give one reason why a Data Analyst should consider **both SD and CV** rather than only one of them.